A normalized local converse for the GI assembly

Author proposal: /root/astra_v14_baseline_theta

7 November 2026

Scope. This author proof, awaiting independent review, derives the precise correlation-to-Gowers-norm implication used by the assembly. It imports the published compact cube constraint, Green–Tao, *Linear equations in primes*, Proposition 11.5, p. 1799, with its definitions in 11.4 and proof in Appendix E. It uses the accepted local increment linearization for polynomial tests. It does not import Proposition 11.2's averaging-domain transition, assume an inverse theorem, or claim to reprove the cube constraint.

1 Available geometric assertion and needed conclusion

Fix an integer $d \geq 1$ and a fixed d-step nilmanifold $M = G/\Gamma$ with a smooth metric. The available cube constraint says there is a compact subset $\Sigma \subseteq M^{\{0,1\}^{d+1}\setminus\{0\}}$ and a continuous map $P : \Sigma \to M$ such that, for every $g \in G$, $x \in M$, $n \in \mathbb{Z}$ and $h \in \mathbb{Z}^{d+1}$, the nonzero vertices $(g^{n+\omega\cdot h}x)_{\omega\neq 0}$ lie in Σ and

$$g^n x = P((g^{n+\omega\cdot h}x)_{\omega\neq 0}).$$

No bound on g or the integers is imposed. In our application G is connected simply connected and at most d-step; these satisfy the published nilmanifold hypothesis.

Theorem 1. *For fixed d, M, L and every $\kappa > 0$ there is $c > 0$ such that every 1-bounded $v : [N] \to \mathbb{C}$ and 1-bounded L-Lipschitz test $F : M \to \mathbb{C}$ satisfying*

$$|\mathbb{E}_{n\in[N]} v(n)F(g^n x)| \geq \kappa$$

obey $\|v\|_{U^{d+1}[N]} \geq c$. The bound is uniform in N, g, x, F. The same conclusion holds for a fixed finite family of rational degree-at-most-d polynomial models, using the exact same-step increment linearization and taking a finite maximum of its bounds.

2 A finite tensor approximation uniform in the test

The algebra generated by coordinate functions that are Lipschitz on M separates points on the compact set Σ, contains constants and is closed under conjugation. Complex Stone–Weierstrass gives, for every fixed continuous F and $\epsilon > 0$, an approximation

$$F(P(z)) = \sum_{a=1}^{A} c_a \prod_{\omega\neq 0} H_{\omega,a}(z_\omega) + R(z), \qquad |R(z)| \leq \epsilon,$$

where the factors are 1-bounded Lipschitz functions. Scaling each factor into the unit disk is absorbed into the finite coefficients c_a; no coefficient is discarded. By Arzela–Ascoli the class

$\|F\|_\infty \le 1$, $\mathrm{Lip}(F) \le L$ is compact in the uniform norm. A finite uniform net, approximated as above, makes A, $\sum_a |c_a|$ and all factor bounds depend only on M, L, ϵ. Choose a net radius $\epsilon/2$ and approximate its members to $\epsilon/2$ to retain the displayed total error.

For each orbit, put $b_{\omega,a}(t) = H_{\omega,a}(g^t x)$. The cube constraint gives

$$F(g^n x) = \sum_a c_a \prod_{\omega \neq 0} b_{\omega,a}(n + \omega \cdot h) + O(\epsilon) \tag{1}$$

for every integer n, h, uniformly. The factors are 1-bounded; their dependence on the orbit will no longer matter in the analytic estimate.

3 Normalized smooth averaging and a uniform Fourier bound

Choose once and for all a smooth $\sigma : \mathbb{R} \to [0,1]$ supported on $[-1,2]$ and equal to one on $[0,1]$. Define

$$S_N = \sum_{h \in \mathbb{Z}} \sigma(h/N), \qquad \mu_N(h) = \sigma(h/N)/S_N.$$

Then $N \le S_N \le 3N + 1 \le 4N$, and μ_N is a probability measure with finite support. Average (1) against the product probability measure $\prod_j \mu_N(h_j)$ on all integer h. The error remains at most ϵ, with no missing normalizing scalar. Thus

$$F(g^n x) = \sum_a c_a B_a(n) + O(\epsilon), \quad B_a(n) = S_N^{-(d+1)} \sum_{h \in \mathbb{Z}^{d+1}} \prod_j \sigma(h_j/N) \prod_{\omega \neq 0} b_{\omega,a}(n + \omega \cdot h).$$

Take $Q = C_d N$ with a fixed integer $C_d > 10(d+2)$, and identify representatives of $\mathbb{Z}/Q\mathbb{Z}$ with a centered interval. The sampled cutoff $w_N(h) = \sigma(h/N)$ has support strictly inside that interval. Write its discrete Fourier expansion as $w_N(h) = \sum_r \widehat{w}_N(r) e(rh/Q)$ with normalized coefficients. There is a bound

$$\sum_{r \in \mathbb{Z}/Q\mathbb{Z}} |\widehat{w}_N(r)| \le C'_d \tag{2}$$

independent of N. To prove it, the zero coefficient and every coefficient are at most $S_N/Q = O_d(1)$. For $r \neq 0$, summation by parts twice gives

$$|1 - e(r/Q)|^2 |\widehat{w}_N(r)| \le Q^{-1} \sum_h |\Delta^2 w_N(h)| \le C/(NQ).$$

The final bound follows from the second derivative of the fixed smooth cutoff, with only $O(N)$ possibly nonzero terms. For the distance $r_* \in [1, Q/2]$ to zero modulo Q, $|1 - e(r/Q)| \ge 4r_*/Q$. Thus $|\widehat{w}_N(r)| \le C_d/r_*^2$ and summation proves (2). Smooth zero extension of σ ensures that the same finite-difference bound holds at its support boundaries.

4 The exact Gowers estimate

Zero-extend v to $\mathbb{Z}/Q\mathbb{Z}$. Extend each bounded integer function $b_{\omega,a}$ to the cyclic group so that it agrees on all integer arguments that occur when $n \in [N]$ and $h_j \in [-N, 2N]$. Those arguments lie between $-(d+1)N$ and $(2d+3)N$; the choice of C_d places the whole interval strictly inside the centered representative range. There is therefore no conflicting cyclic value. The normalized correlation with B_a is exactly

$$\frac{Q}{N} \left(\frac{Q}{S_N}\right)^{d+1} \mathbb{E}_{n,h \in \mathbb{Z}/Q\mathbb{Z}} v(n) \prod_j w_N(h_j) \prod_{\omega \neq 0} b_{\omega,a}(n + \omega \cdot h).$$

The prefactor is bounded above by a constant depending only on d. This identity comes from an all-integer smooth average; no hard box average is replaced by a larger summation domain.

Expand the product of cutoffs by Fourier series. The sum of absolute values of its coefficients is at most $(C'_d)^{d+1}$. In each term, the factor $e(\sum_j r_j h_j/Q)$ can be absorbed into the nonzero vertex functions. For each j, choose an index $k \neq j$, which exists since $d+1 \geq 2$, and use $h_j = (n+h_j+h_k)-(n+h_k)$. Multiply the functions at vertices e_j+e_k and e_k by the corresponding opposite linear phases. Their absolute bounds remain one. The Gowers–Cauchy–Schwarz inequality for the full $(d+1)$-cube gives the absolute bound $\|v\|_{U^{d+1}(\mathbb{Z}/Q\mathbb{Z})}$ for every such term. Complex conjugations are absorbed into the vertex functions, leaving the norm and absolute bounds unchanged. Hence

$$|\mathbb{E}_{[N]} v B_a| \leq K_d \|v\|_{U^{d+1}(\mathbb{Z}/Q\mathbb{Z})} \leq K_d \|v\|_{U^{d+1}[N]}.$$

The last inequality uses the prescribed normalized interval norm: the cyclic norm is the interval norm multiplied by the cyclic norm of the interval indicator, which is at most one. If necessary enlarge C_d also to exceed 2^{d+1} so this single ambient group realizes that interval normalization.

Combining the finite tensor approximation with this estimate yields

$$|\mathbb{E}_{[N]} v(n) F(g^n x)| \leq K_d \Big(\sum_a |c_a| \Big) \|v\|_{U^{d+1}[N]} + \epsilon.$$

Choose $\epsilon < \kappa/2$. This proves Theorem 1 with a positive bound depending only on d, M, L, κ. A fixed bound other than one for v or F is handled by dividing by that bound first. This proof is complex-valued throughout; it does not need a real-valued dual-norm convention.

5 Polynomial input and source comparison

For the actual assembly input, a scalar polynomial nilsequence has degree $d = s-2 \geq 1$ on fixed rational geometry. The accepted increment linearization001 represents it exactly on a fixed group of step at most d, with a fixed Lipschitz multiplier and unchanged norm bound. Applying the theorem there and taking finite maxima over a finite family gives precisely the positive normalized $U^{s-1}[N]$ norm required by the last induction step. This is a converse implication and has no inverse-theorem premise.

The literal published proof of 11.2 on p. 1801 defines an average over $h \in [N]^{d+1}$ with $\sigma(h_j/N)$, while $\sigma = 1$ on $[0,1]$. Its later passage from (11.6) to (11.7) widens the shift domain to a cyclic group with the same cutoff supported on $[-1,2]$. That changes the set of summands. The local proof above instead starts with the normalized all-integer smooth average and tracks its exact cyclic prefactor. Whether an earlier source correction already supplied this interpretation has not been established. The observation is a bounded proof-transition issue, not a counterexample to Proposition 11.2 or the converse theorem.

The related-use scope is the normalization of the 11.2 average, its dual-norm estimate, Corollary 11.6, and the present GI converse application. Published 11.5 remains an explicit unchanged supplier whose source-level application must be independently accepted. No claim of whole-source acceptance of Appendix E or of the full GI assembly follows from this local analytic proof.